

wavenumber modes (and stronger damping of intermediate modes with the fourth-order viscosity), with very close results for all tested time integration methods. As illustrated in Figure 12 for the SE22 scheme (qualitatively similar results are obtained using SE12 or SL-SI-SETTLS), the sixth-order viscosity produces a better approximation to the reference solution provided by [34].

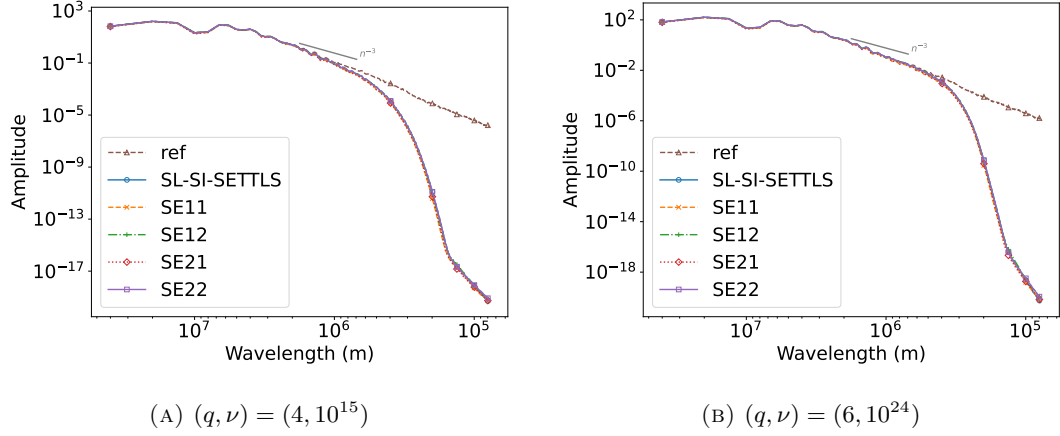


FIGURE 11. Kinetic energy spectra after six days, with $\Delta t = 240s$ (and $\Delta t = 2s$ for the reference solution), truncated at resolution $M = 512$, for the unstable jet case with artificial viscosity.

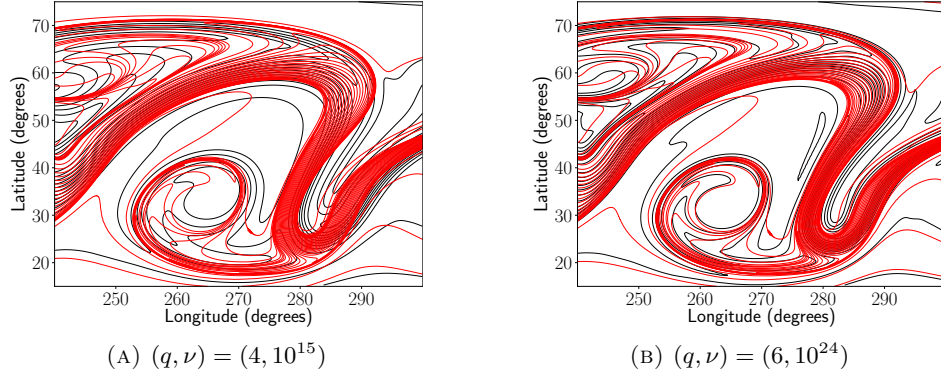


FIGURE 12. Contour lines of the potential vorticity field after six days of the unstable jet test case with artificial viscosity, using the SE22 scheme and $\Delta t = 240s$ (black lines), compared to the reference solution provided by [34] (red lines). Contour levels are separated by $0.2\Omega/\bar{h}$.

By comparing these results with the simulations without viscosity (Figure 9), we observe that SE22 with $\Delta t = 120s$, $\nu = 0$ is qualitatively superior to the simulation with $\Delta t = 240s$ and $(q, \nu) = (6, 10^{24})$, but the latter outperforms the simulation with $\Delta t = 960s$, $\nu = 0$, which presents several small-scale oscillations despite a relatively stable simulation compared to other time integration

schemes. On the other hand, SL-SI-SETTLS and SE12 clearly benefit from the use of artificial viscosity, with considerably more stable and accurate results compared to the inviscid case with both $\Delta t = 120\text{s}$ and $\Delta t = 960\text{s}$ (in the case of SL-SI-SETTLS) or $\Delta t = 960\text{s}$ (in the case of SE12).

8. CONCLUSION

Semi-Lagrangian exponential methods, proposed by [26], have a potential for application in atmospheric circulation models, since they have enhanced stability compared to Eulerian exponential schemes and, contrary to traditional semi-Lagrangian methods, are able to integrate the linear terms very accurately; therefore, this class of time stepping scheme is more suitable for the integration of advection-dominated models containing stiff linear terms. The work presented here aimed to extend [26] in several aspects: first, we identified and explained the accuracy limitations of their methods, showing that the combination of exponential and interpolation operators adopted in the discretization of the linear term, which corresponds to a first-order discretization of the integration factor, produces a first-order leading term in the truncation error. Second, inspired by this initial study, we proposed a new method, with an alternative discretization of the linear term, leading to an effective second-order accuracy. Also, we conducted a detailed stability study comparing several Eulerian and semi-Lagrangian methods, both through a linear stability analysis and an empirical simulation-based stability study, which was required because the standard stability analysis is not able to provide detailed insights in the stability of semi-Lagrangian exponential schemes. Finally, while [26] developed their work in the context of the SWE on the bi-periodic plane, we applied the schemes to the integration of the SWE on the rotating sphere, thus providing results that are more closely related to real-world operational atmospheric circulation models.

Promising results were obtained using the proposed second-order semi-Lagrangian exponential scheme. Its theoretical convergence order was confirmed in several benchmark test cases, and, considering the especially challenging unstable jet simulation, it proved to be more stable and accurate than Eulerian exponential schemes, other semi-Lagrangian exponential schemes and the well-established second-order SL-SI-SETTLS, even without the use of artificial viscosity approaches, which is a consequence of the combined accurate integration of the stiff linear terms and the semi-Lagrangian treatment of the advection. The latter approach largely extends the range of stable time step sizes compared to a second-order Eulerian exponential method, maintaining approximately the same accuracy, while the former leads to larger accuracy, compared to SL-SI-SETTLS, in simulations dominated by linear processes; however, our proposed scheme is outperformed by SL-SI-SETTLS (if stable) when nonlinear effects are more important. In terms of computational cost, a drawback of the proposed method is its higher complexity compared to the previously formulated first-order semi-Lagrangian exponential schemes, as a consequence of the splitting of the exponential operator applied to the linear term, which is required to achieve second-order accuracy. We highlight, however, that this larger cost can be mitigated by a straightforward computation of matrix exponentials, by combining a spectral discretization with a convenient arrangement of the terms of the governing equations, and by precomputing these exponential terms, which is possible in simulations with a constant time step size. Also, the observed wall-clock times are comparable to those of SL-SI-SETTLS, which is used in operational atmospheric circulation models; therefore, our proposed method seems to be a competitive candidate for operational applications.

Important aspects still need to be considered in future work for further improvement of semi-Lagrangian exponential methods. As illustrated in the numerical simulations, these schemes presented unexpected unstable behaviors in a specific range of time step sizes, which can possibly be

caused by badly behaved exponential functions; even if these issues have been easily avoided by using high-order artificial viscosity, with little damage to the accuracy of the numerical solution, a more detailed understanding needs to be achieved. Also, higher-order semi-Lagrangian exponential schemes can be formulated to achieve improved accuracy in simulations dominated by nonlinearities; from the truncation error analyses conducted here, they could probably be obtained in a quite straightforward way following the derivation of higher-order Eulerian exponential methods by [8] and using higher-order approximations to the integral involved in the integration factor term; however, the hypothesis on the commutation of the linear operator along the Lagrangian trajectories, which was proved here to introduce a second-order error, should be carefully studied. Moreover, the discussion on the derivation of linear stability analyses of semi-Lagrangian exponential schemes combined with spectral methods, conducted in Section 6.4, suggests that more intricate studies should be necessary to fully understand the relation between spectral transforms, matrix exponentiation, and spatial interpolation, and the influence on accuracy properties of semi-Lagrangian exponential methods, which would certainly contribute to further developments of this class of methods. Finally, applications to more complex atmospheric models would be a natural and challenging extension of this work.

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DATA AVAILABILITY STATEMENT

The research code, benchmark tests and postprocessing scripts used in this article are available in the repository https://gitlab.inria.fr/sweet/sweet/-/tree/paper_2nd_order_SLETDRK/benchmarks_sphere/paper_2nd_order_sletdrk [4].

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